Diseases of the Liver & Gallbladder

NURA 340 Pathophysiology | Final Exam Lecture Study Guide

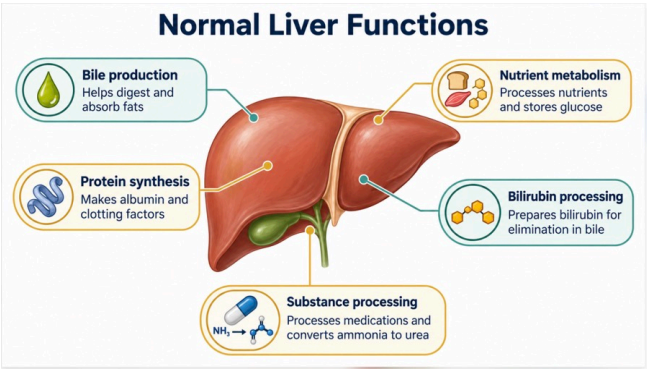
Cholelithiasis, cholecystitis, cirrhosis, hepatitis A/B/C, liver labs

Lecture roadmap: Normal liver functions → Bile and bilirubin pathway → Cholelithiasis → Cholecystitis → Intro to liver disease and stages of liver failure → Cirrhosis (two mechanisms) → Why cirrhosis causes bleeding → Hepatitis A, B, C → Hepatitis comparison → Liver and biliary labs

How the instructor wants you to think: Ask **what does this organ normally do, and what happens when it can't?** That connects gallstones, pain, and jaundice, and links liver dysfunction to **edema, bleeding, and mental status changes**. For cirrhosis, tie every finding to **(1) lost liver function** or **(2) portal hypertension**, or both. Nursing interventions are not tested.

How to read this guide: Everything without a tag comes from the lecture slides and transcript. Anything else is tagged: ADDED information not in the lecture; CLARIFIED a lecture point explained further or a garbled transcript line cleaned up; CORRECTED an error in the slides or transcript. Every tag is listed in the notes at the end.

Normal Liver Functions



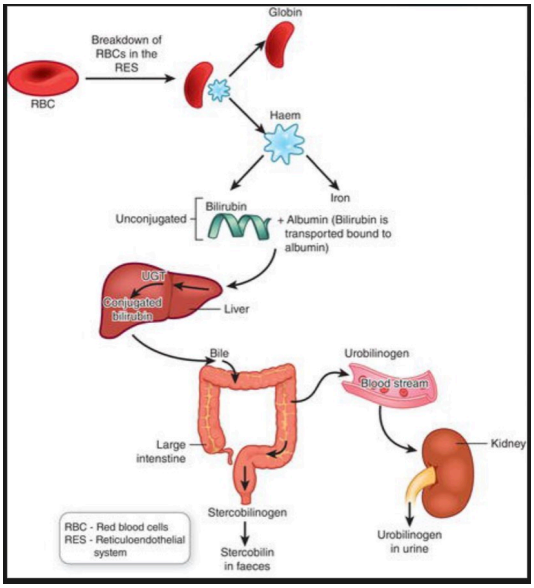
From the lecture slides.

Function	What it does	If it fails, expect…
Nutrient metabolism	Processes nutrients from the GI tract; stores glucose as **glycogen** and releases it when needed; processes fat and protein	Malnutrition, muscle loss
Bile production	Makes **bile** to digest and absorb **fat**	Poor fat digestion
Bilirubin processing	**Processes** (doesn't make) **bilirubin**, the waste product from normal RBC breakdown, and sends it into bile for elimination in feces	**Jaundice**, dark urine, pale stool
Protein synthesis	Makes **albumin** (keeps fluid in the bloodstream) and many **clotting factors**	**Edema, ascites**; **bleeding, bruising**
Substance processing	Processes **medications**; filters **ammonia** (from protein metabolism) and converts it to urea	Drug buildup; **confusion (hepatic encephalopathy)**

Bile and Bilirubin Pathway ADDED

The slide shows this diagram, but the recording skips this part of the lecture, so the step-by-step explanation below is mine. You need it to understand jaundice, dark urine, and pale stool.

1. Old **RBCs are broken down** (mostly in the spleen). Heme → **unconjugated bilirubin** + iron.
2. Unconjugated bilirubin is **not water-soluble**, so it travels to the liver **bound to albumin**.
3. The **liver conjugates** it (enzyme UGT), making **conjugated bilirubin**, which **is water-soluble**.
4. Conjugated bilirubin goes into **bile** → stored in the gallbladder → released into the intestine.
5. In the intestine, bacteria convert it to urobilinogen. Most becomes **stercobilin**, which gives **stool its brown color**. A little is reabsorbed and excreted by the kidneys as **urobilinogen** in urine.



From the lecture slides (Clinical Guide to Paediatrics, 2022).

Why does a blocked bile duct cause dark urine and pale stool? (Instructor emphasis: make sure you understand this.)
Conjugated bilirubin can't reach the intestine, so it **backs up into the blood → jaundice**. Because it's **water-soluble**, the kidneys filter it into the urine → **dark (tea-colored) urine**. Less bilirubin reaches the gut → less stercobilin → **pale, clay-colored stool**. Normally, bilirubin is **not** found in urine.

Gallbladder Disease: Overview

The gallbladder **stores and concentrates bile** until it's needed. It connects to the biliary system through the **cystic duct**. Bile from the liver and gallbladder reaches the intestine through the **common bile duct**.

Cholelithiasis (Gallstones)

Cholelithiasis simply means **gallstones are present**. Many people have stones or tiny crystals and never know it, because nothing is blocked. **Location matters**: a stone sitting in the gallbladder may cause no problem, while a stone blocking a duct interferes with bile flow.

Etiology	Bile contains **cholesterol and bile salts**. If there's more cholesterol than can stay dissolved, crystals form. **Most gallstones are mainly cholesterol.** Stones are more likely when the gallbladder **doesn't empty well** and bile sits longer (**stasis**). Risk factors: **increasing age**, **rapid weight loss**, **pregnancy** (hormones change bile composition and gallbladder emptying), certain **metabolic conditions**.
Pathophysiology	Crystals → combine into a **stone** → the stone either stays put (no problem) or **moves into a duct**: • **Cystic duct blocked:** the gallbladder contracts against the blockage → pressure builds → **pain**. If it persists → the wall inflames → **cholecystitis**. • **Common bile duct blocked:** bile from the **liver** can't reach the intestine → bilirubin backs up → **jaundice**.
Manifestations	• Many people have **no symptoms**. • Obstruction → **right upper quadrant (RUQ) or epigastric pain**, which can be **referred to the right shoulder or back**; **nausea and vomiting**. • **Worse after a fatty meal**: fat in the small intestine signals the gallbladder to contract and release bile. If the outlet is blocked, the contraction pushes against the stone → more pressure → more pain. • **Common bile duct** stone → **jaundice**, **dark urine** (bilirubin in urine), pale stool. • A stone blocking **only the cystic duct** usually does **not** cause jaundice, because bile from the liver can still reach the intestine.
Diagnosis	**Ultrasound** is usually the first imaging test. It shows stones and signs of inflammation such as a **thickened gallbladder wall**. Blood tests: elevated **bilirubin** or **alkaline phosphatase (ALP)** suggests a bile duct problem; interpret with symptoms and imaging.
Impact on health	Depends on **where the stone is**: • Cystic duct → **cholecystitis** • Common bile duct → obstructed bile flow, possible **infection of the bile ducts** • Near where the bile and pancreatic ducts empty into the intestine → blocks pancreatic drainage → **gallstone pancreatitis** (less common)

Management	No symptoms → often **no treatment**. Symptomatic stones → **cholecystectomy** (gallbladder removal) is common; medications for pain or nausea.

Think it through: What changes after the gallbladder is removed?
The liver keeps making bile, but bile now flows **straight into the intestine without being stored**. After a **fatty or heavy meal**, there is no gallbladder to hold and meter out bile. A rush of bile into the intestine can cause **urgent, bile-heavy diarrhea**. People are encouraged to eat a **lower-fat diet**. CLARIFIED The lecturer called this "dumping." See the notes page for how this differs from dumping syndrome.

Cholecystitis

Cholecystitis = inflammation of the gallbladder (*-itis*).

Etiology	• About **90%** are caused by a **gallstone blocking the cystic duct** (calculous). • **Acalculous cholecystitis** = **without a stone** (*a-* = without, *calculus* = stone), mainly in **critically ill** people.
Pathophysiology	Cystic duct stays blocked → **bile trapped** → **pressure rises** → wall stretches → **inflammation** → swelling impairs **blood flow** to the wall → possible **necrosis**. **Bacterial infection** can develop, though inflammation doesn't always start with infection. Repeated episodes → **chronic inflammation and scarring**.
Manifestations	**RUQ pain and tenderness**, nausea or vomiting, **fever** (inflammatory response or infection). The pain is **more persistent** than a brief episode of biliary pain, because the wall itself is inflamed. It can be **referred to the right shoulder or back**.
Diagnosis	Symptoms + physical findings + labs + imaging. ↑**WBC or CRP** supports inflammation but doesn't show where. **Ultrasound** (usually first after labs): gallstones, **thickened wall**, **fluid around the gallbladder**. **Bilirubin may be normal** in uncomplicated cholecystitis; a **significant rise** suggests obstruction **beyond the cystic duct** (common bile duct).
Impact on health	Impaired blood flow → **necrosis** → **perforation** → bile or infected material leaks into the abdomen → **peritonitis**.

Management	Usually **gallbladder removal**; **antibiotics** if infection is suspected; a separate **endoscopic procedure** may remove a stone stuck in the common bile duct. Focus on **why the location of the obstruction matters**, not treatment details.

Pattern to recognize: Cholecystitis follows the **same progression as appendicitis**: **obstruction → pressure → inflammation → compromised perfusion → tissue death → rupture → peritonitis**.

Introduction to Liver Disease

- **Hepatitis** = inflammation of the liver. Depending on the cause, it may **resolve** or **persist**.
- When liver injury persists, normal healing produces **fibrosis** (scar tissue). Extensive scarring and structural change → **cirrhosis**.
- Connect each local injury to **what the liver can no longer do** and to **what happens when blood has trouble moving through it**.

Stages of Liver Failure

You don't need to stage patients, but understand the progression:

1. **Compensated**: part of the liver is damaged, but the body works around it for a time.
2. **Decompensated**: the body can no longer compensate, and complications appear.
3. **End stage**

Cirrhosis

Healthy liver tissue is **permanently replaced by scar tissue** after long-term damage and inflammation.

The two mechanisms (instructor emphasis):

1. Loss of normal liver function	2. Portal hypertension
Less ability to make **albumin** and **clotting factors**, process **bilirubin**, and clear **ammonia** (and drugs)	Blood from the GI tract reaches the liver through the **portal vein**. Scarring makes the path **more resistant** → **pressure rises** in the portal circulation
→ **edema/ascites** (low albumin), **bleeding/bruising** (low clotting factors), **jaundice**, **hepatic encephalopathy** (ammonia)	→ **ascites** (fluid pushed out of the circulation), **splenomegaly** (→ low platelets), **varices**

The two often occur together. Ascites is a good example: portal hypertension pushes fluid out, and **low albumin** makes it worse.

Etiology	Chronic liver injury from: • Long-term **alcohol-associated liver disease** • Chronic **hepatitis B or C** • **MASLD**: metabolic dysfunction-associated steatotic (fatty) liver disease • Autoimmune disease, inherited conditions, chronic bile duct problems The common feature: **ongoing injury + repeated attempts at repair = scar tissue**.
Pathophysiology	Repeated injury → inflammation and healing → **fibrosis**. The liver forms **regenerative nodules** that distort its structure. This interferes with **blood flow** (portal hypertension) and impairs **normal functions**. Early on, the liver **compensates**. As function declines and portal hypertension worsens, complications become apparent.

Manifestations	**Early: vague** — fatigue, decreased appetite, weakness. **Later**, tied to the mechanisms: • ↓ albumin → **edema** • ↓ bilirubin processing → **jaundice** • ↓ clotting factors + ↓ platelets → **bleeding, bruising** • Portal hypertension → **ascites** (excess fluid in the peritoneal cavity), **splenomegaly**, **varices** • ↑ **ammonia** reaching the brain → **confusion, hepatic encephalopathy** Don't memorize a long list. Link each finding to **impaired function**, **portal pressure**, or **both**.
Diagnosis	History, exam, labs, imaging; tests of **liver stiffness**; **biopsy** if uncertain. • **AST/ALT** rise with liver **cell injury** but **don't show how well the liver works**. In advanced cirrhosis they may be only mildly elevated or **even normal**. • **Albumin and INR** show the liver's ability to **make proteins**. • **Bilirubin** shows how well it's processed and cleared.
Impact on health	Widespread: **ascites and edema** (fluid distribution); **varices** → serious **GI bleeding**, made worse by poor clotting; **hepatic encephalopathy** (attention, sleep, behavior, mental status); **malnutrition and muscle loss**. These can occur even when some liver labs don't look dramatically abnormal.
Management	Address the **underlying cause** and the **complications**; comfort and quality of life; some people need a **liver transplant**.

Discussion Question #3: Why is a person with cirrhosis at risk for bleeding?

Two related questions:

1. **Can the body form a clot?** The liver makes most **clotting factors**; with cirrhosis it makes fewer. **Portal hypertension** → **enlarged spleen**, which holds more platelets → **fewer circulating platelets**.
2. **Is there a vessel likely to rupture?** Portal hypertension creates **varices**. If one ruptures, bleeding is heavy, and the body can't clot well.

Clotting in cirrhosis is complex, and some patients actually develop clots. For this course, focus on the links among **clotting factor production, platelets, portal hypertension, and varices**.

Viral Hepatitis

All three viruses cause **liver inflammation**, and their **symptoms overlap**: fatigue, nausea, decreased appetite, RUQ discomfort, ± jaundice, dark urine, pale stools. What separates them is **how they spread**, **whether they become chronic**, and **whether there's a vaccine**.

Hepatitis A

- **Spread: fecal-oral route**, through contaminated food or water or close contact with an infected person.
- The virus reaches the liver → inflammation → disrupts liver function and bilirubin handling.
- Symptoms: fatigue, nausea, decreased appetite, RUQ discomfort; possibly jaundice, dark urine, pale stools. **Young children may have few or no symptoms.**
- Diagnosis: markers of liver injury (↑**AST/ALT**) and **antibodies** showing recent hepatitis A infection.
- **Acute only. It does NOT become chronic.** Most people recover. Severe injury is possible in people with **existing liver disease**.
- Treatment is **supportive** while the liver recovers. A **vaccine** exists.

Instructor emphasis: "Hepatitis A acute infection does not become chronic." Know the **route (fecal-oral)** and that it is **never chronic**.

Hepatitis B

- **Spread: infected blood and certain body fluids** — sexual contact, shared injection equipment, needle sticks, and from a pregnant person to the infant **around birth (perinatal)**.
- **Highly infectious**: a tiny amount of blood is enough, and the virus stays infectious on surfaces for **at least 7 days**, even when blood isn't visible. This is why **healthcare workers** (and everyone) should be vaccinated.
- **Patho:** the virus infects liver cells, and the **immune response** against it causes inflammation and liver injury. Some people clear it; in others it becomes **chronic**.
- **Age matters: infants are much more likely than adults to develop chronic infection**. The younger the person is at infection, the more likely lifelong liver disease, cirrhosis, liver failure, liver cancer, or transplant. Adults can also become chronic.
- Symptoms resemble other hepatitis. **Many have no obvious symptoms**, especially with chronic infection, so a person can be chronically infected without feeling ill.
- **Diagnosis:** antigens and antibodies. **Surface antigen (HBsAg)** = **current infection**. **Surface antibody (anti-HBs)** = **immunity** from vaccination or recovery. You don't need every marker combination. Know that testing distinguishes **current infection** from **immunity or past infection**.

- **Chronic course:** ongoing inflammation → **fibrosis** → **cirrhosis** → **liver failure** → **liver cancer**.
- **Vaccine**: very well tolerated with lots of safety data. The instructor noted a worrying drop in parents choosing it for newborns; nurses play a key role in patient education. Adults without immunity (checked with a **titer**) complete the vaccine series.
- **Antivirals** can **suppress** chronic infection and reduce progression.
- Infants born to a person with active hepatitis B also get **HBIG** (hepatitis B immune globulin) shortly after birth, for immediate antibody protection (passive immunity). This is extra information, not tested.

Hepatitis C

- **Spread: primarily blood** — shared injection equipment, inadequately sterilized medical equipment; also **pregnant person** → **infant**.
- The virus stays in the liver and causes **ongoing inflammation**. In **more than half** of people, the infection becomes **chronic**. Repeated inflammation and healing → **fibrosis and cirrhosis**. It is the virus most associated with long-term chronic liver disease.
- Often **no noticeable symptoms for years**; symptoms that do appear are nonspecific (fatigue, abdominal discomfort). **No jaundice or pain does not mean the liver is unaffected.**
- **Diagnosis: antibody test** first. A positive result means **exposure at some point**, not necessarily current infection. An **HCV RNA test** looks for the virus itself and confirms **current infection**.
- Chronic → cirrhosis, portal hypertension, ascites, **liver cancer**.
- **Direct-acting antivirals can cure most infections. No vaccine.**

Hepatitis Comparison

Feature	Hepatitis A	Hepatitis B	Hepatitis C
Transmission	**Fecal-oral**	**Blood and certain body fluids**; sexual, perinatal	**Primarily blood**
Course	**Acute only**; never chronic	Acute **or** chronic (infants are at highest risk of chronic)	**Often chronic** (more than half)
Vaccine?	**Yes**	**Yes**	**No**
Treatment	Supportive care	Antivirals **suppress** chronic infection	Antivirals **cure** most infections

Feature	Hepatitis A	Hepatitis B	Hepatitis C
Key test point	Antibodies to recent infection	HBsAg = current infection; anti-HBs = immune	Antibody = exposure; RNA = current infection

Memory aid ADDED **A** = **A**limentary (fecal-oral), **A**cute only. **B** = **B**lood and **B**ody fluids, **B**abies become chronic. **C** = **C**hronic, **C**urable, **C**an't vaccinate.

Liver and Biliary Labs

For each lab, ask: is this telling me about **liver cell injury**, **bile flow**, or the liver's **ability to make proteins**?

Lab	What it assesses	Pathophysiology connection
AST and ALT	**Liver cell injury**	Rise when injured cells release these enzymes (e.g., hepatitis). They **do not measure function**; unlike GFR for the kidneys, they can't tell how well the liver works.
ALP and GGT	**Bile flow** and biliary injury	Rise when bile flow is impaired (e.g., a stone in a bile duct). ALP also comes from **bone**; **GGT** helps show whether a high ALP is from the liver or biliary system.
Bilirubin	Processing and excretion of bilirubin	↑ → **jaundice**; blocked bile flow → **dark urine** and **pale stool**
Albumin	**Protein synthesis**	↓ → fluid leaks out of the vessels (third spacing) → **edema, ascites**. Can be low for other reasons too (e.g., kidney).
PT / INR	**Clotting factor synthesis**	Prolonged PT / ↑INR → reduced clotting factor production. Also affected by some medications and **vitamin K deficiency**.
Platelets	Cells involved in clotting	↓ with **portal hypertension** → **enlarged spleen** holds platelets

Putting it together: A person with **hepatitis** may have **high AST/ALT** (cells are being injured right now). A person with **advanced cirrhosis** may have **normal or mildly high AST/ALT** but **low albumin and high INR**, a sign the liver can't make proteins. No single lab tells the whole story; connect labs to what you see: **jaundice, edema, ascites, bruising**.

Quick Review

Disorder	Core mechanism	Hallmark findings	Key complications
Cholelithiasis	Too much cholesterol in bile + stasis → stones	Often none; RUQ/epigastric pain after fatty meals → R shoulder; N/V	Cholecystitis, bile duct infection, gallstone pancreatitis
CBD stone	Bile from the liver blocked	Jaundice, dark urine, pale stool; ↑bilirubin, ↑ALP	Bile duct infection, pancreatitis
Cholecystitis	Cystic duct blocked (90%) → pressure → inflammation; acalculous in critical illness	Persistent RUQ pain/tenderness, fever, ↑WBC/CRP; bilirubin often normal	Necrosis, perforation, peritonitis
Cirrhosis	Permanent scarring: ↓ function + portal HTN	Early vague; later edema, ascites, jaundice, bruising, confusion	Variceal bleeding, encephalopathy, malnutrition, liver failure
Hep A	Fecal-oral virus	Acute hepatitis symptoms; kids may have none	Never chronic
Hep B	Blood/body fluid virus; immune response injures the liver	Often silent when chronic	Chronic → cirrhosis, liver cancer (especially if infected as an infant)
Hep C	Bloodborne virus; persistent inflammation	Silent for years	Chronic in more than half → cirrhosis, liver cancer

Self-Check ADDED

1. Why does a stone in the cystic duct usually not cause jaundice, while a common bile duct stone does?
   With a cystic duct stone, bile from the liver can still reach the intestine through the common bile duct. A common bile duct stone blocks it, so bilirubin backs up into the blood.
2. Why is gallbladder pain worse after a fatty meal?
   Fat in the intestine signals the gallbladder to contract. Contracting against a blocked outlet raises pressure and pain.
3. Why does gallbladder pain show up in the right shoulder?
   Referred pain: visceral pain is perceived at a different location because of shared nerve pathways.
4. What is acalculous cholecystitis, and who gets it?
   Gallbladder inflammation without a stone, mainly in critically ill people.
5. Name the two mechanisms of cirrhosis and one finding from each.
   Loss of function (low albumin → edema; low clotting factors → bleeding; ammonia → encephalopathy) and portal hypertension (ascites, splenomegaly, varices).
6. Why can a patient with advanced cirrhosis have normal AST/ALT?
   These enzymes reflect active cell injury, not function. With few working cells left, there may be little enzyme to release.
7. Why is a cirrhotic patient's platelet count low?
   Portal hypertension enlarges the spleen, which holds more platelets.
8. Which hepatitis is spread by the fecal-oral route and never becomes chronic?
   Hepatitis A.
9. Why is the hepatitis B vaccine recommended for newborns?
   Infants infected with HBV are much more likely to develop chronic infection, which leads to cirrhosis, liver failure, and liver cancer.
10. A patient's hepatitis C antibody is positive. Do they have hepatitis C now?
    Not necessarily. The antibody shows exposure; an HCV RNA test confirms current infection.
11. Which hepatitis has no vaccine but is usually curable?
    Hepatitis C.
12. Which lab best reflects the liver's ability to make clotting factors?
    PT/INR.

Notes on Sources, Additions & Clarifications

This guide was written from the lecture slides (Diseases of the Liver and Gallbladder, NURA 340), including the hepatitis comparison and liver lab tables, and the organized lecture transcript.

- ADDED **Bile and bilirubin pathway.** The slides include the pathway diagram (slides 3–4), but the transcript jumps from 2:35 to 5:12, so the explanation of these slides isn't recorded. The step-by-step explanation is mine, written from standard A&P and matched to the slide diagram. The "dark urine, pale stool" explanation in the box is from the lecture (the labs section).
- CLARIFIED **"Dumping" after cholecystectomy.** The lecturer describes urgent, bile-heavy diarrhea after a fatty meal when there's no gallbladder and calls it "dumping." The mechanism described is correct. Be aware that **dumping syndrome** usually refers to something different: rapid emptying of stomach contents into the small intestine after **stomach** surgery. If an exam question says "dumping syndrome," it most likely means the gastric version.
- CLARIFIED **Acalculous cholecystitis.** The transcript renders it as "a calculus cholecystitis." The correct term is *acalculous* (a- = without, calculus = stone).
- CLARIFIED **Cirrhosis and resistance.** At one point the transcript says cirrhotic tissue causes "less resistance," then corrects itself to "more resistance." Scarring **increases** resistance to portal blood flow, which is what causes portal hypertension.
- CLARIFIED **Liver functions table and HBsAg/anti-HBs abbreviations.** The functions and failure findings come from the lecture. The abbreviations HBsAg and anti-HBs are standard names for the "surface antigen" and "surface antibody" the lecturer describes.
- ADDED **Hepatitis A/B/C memory aid** and **Self-Check questions** are my own.
- HBIG and the hepatitis B vaccine history are included as context; the lecturer said these are not tested in this course.